

MATHEMATICS 1

40 MINUTES

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but **do not open this question paper until you are ready to begin.**

This paper contains **27 multiple-choice questions**. There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all of the questions. Each question is worth one mark.

For each question, choose the one option you consider correct. If you make a mistake, erase thoroughly and try again.

You must complete the paper within the time limit.

You can use the question paper for rough working. Dictionaries and calculators are **NOT permitted**.

Please wait until you are ready before turning this page.

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Mathematics 1

- 1 A dataset of $2n$ numbers is partitioned into two groups of n numbers. Two numbers are then removed from the dataset.

The table below summarises the number of values and the mean for each group, as well as for the removed and remaining values:

Values	Number of values	Mean
First group	n	a
Second group	n	b
Removed values	2	c
Remaining values	$2n-2$	m

Which of the following is a correct expression for n in terms of a , b , c , and m ?

- A $\frac{2(c-m)}{a+b-2m}$
- B $\frac{2(c-m)}{a+b-m}$
- C $\frac{c-m}{a+b-2m}$
- D $\frac{2(m-c)}{a+b-2m}$
- E $\frac{2(c+m)}{a+b-2m}$
- F $\frac{2(c-m)}{a+b+2m}$

- 2 A rectangle is divided into four smaller non-overlapping rectangles by two straight lines parallel to its sides, as shown in the diagram.

The area of the top-left rectangle is 28 cm^2 .

The area of the bottom-left rectangle is 42 cm^2 .

The combined area of the two rectangles on the right is 65 cm^2 .

What is the area of the bottom-right rectangle?

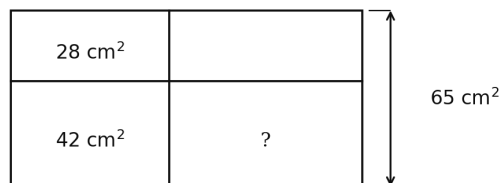


Diagram not to scale

- A** 26 cm^2
- B** 33 cm^2
- C** 39 cm^2
- D** 45 cm^2
- E** 51 cm^2
- F** 81 cm^2

- 3 The curve with equation

$$y = \log_2(x)$$

meets the curve with equation

$$y = 3 - \log_2(k - x)$$

where k is a positive constant, at two distinct points.

The x -coordinates of these two points are p and q , where $q > p$.

Given that $q - p = 2$, what is the value of k ?

- A** 4
- B** $2\sqrt{5}$
- C** $4\sqrt{2}$
- D** 6
- E** 8
- F** $2\sqrt{10}$

- 4 The positive integers are written in a grid with 6 columns, numbered 1 to 6 from left to right. The first row contains the integers 1 to 6, placed in columns 1 to 6 respectively. The second row contains the integers 7 to 12, placed in columns 6 down to 1 respectively. The third row contains the integers 13 to 18, placed in columns 1 to 6 respectively. This alternating pattern continues indefinitely.

In which column is the integer 250?

- A 1
- B 2
- C 3
- D 4
- E 5
- F 6

- 5 A box contains 5 fair six-sided dice and 3 biased six-sided dice.

For each biased die, the probability of rolling a 6 is $\frac{1}{2}$, and the numbers 1 to 5 are equally likely.

A die is chosen at random from the box and is rolled twice.

Given that the first roll shows a 6, what is the probability that the second roll also shows a 6?

- A $\frac{1}{9}$
- B $\frac{1}{6}$
- C $\frac{7}{24}$
- D $\frac{1}{3}$
- E $\frac{8}{21}$
- F $\frac{1}{2}$
- G $\frac{9}{14}$

- 6 A car travels a distance of D miles. The fuel consumption of the car is F litres per 100 kilometres.

Given that 5 miles is equivalent to 8 kilometres, which of the following expressions represents the total volume of fuel consumed, in litres?

- A $\frac{2DF}{125}$
- B $\frac{DF}{160}$
- C $\frac{8DF}{5}$
- D $\frac{5DF}{8}$
- E $160DF$
- F $\frac{125DF}{2}$

- 7 A straight line L passes through the origin and the point $(4, 12)$.

A point P in the first quadrant is at a distance of 18 units from the origin. The line segment connecting the origin to P makes an angle of 30° with the positive x -axis.

What is the sum of the gradient of L and the y -coordinate of P ?

- A 9
- B 12
- C 21
- D $\frac{28}{3}$
- E $3 + 9\sqrt{3}$
- F $\frac{1}{3} + 9\sqrt{3}$

- 8 Two fair, six-sided dice are thrown. Given that the sum of the numbers shown is at least 9, what is the probability that the two numbers are equal?

- A $\frac{1}{6}$
- B $\frac{1}{5}$
- C $\frac{5}{18}$
- D $\frac{1}{3}$
- E $\frac{3}{5}$

- 9** Water flows from a cylindrical pipe with an internal diameter of 200 mm into an open rectangular tank. The base of the tank is a square with side length 10 cm. The water flows through the pipe at a constant velocity of 1.2 m/s.

What is the rate of increase of the depth of the water in the tank, in cm/min?

- A** 2π
- B** 8π
- C** 720π
- D** 7200π
- E** 28800π
- F** 72000π

- 10** A particle undergoes symmetric, periodic oscillation about a central point. The maximum deviation of the particle from the central point is 10 cm, and the duration of one complete cycle is 0.50 s.

What is the total distance traveled by the particle in a time interval of 3.0 s?

- A** 0 cm
- B** 40 cm
- C** 60 cm
- D** 120 cm
- E** 240 cm

- 11** A solid metal sphere of diameter 6 cm and a solid metal cylinder of diameter 8 cm and height 3 cm are melted down. The molten metal is used to cast a number of identical solid cones, each with diameter 2 cm and height 3 cm.

Assuming no metal is lost, how many complete cones can be made?

- A** 28
- B** 36
- C** 48
- D** 75
- E** 84
- F** 120

- 12** A number of coin flips, N , is chosen with equal probability from the set $\{2, 3, 4\}$. A fair coin is then flipped N times.

What is the probability that exactly two Heads are observed?

- A** $\frac{1}{4}$
- B** $\frac{5}{14}$
- C** $\frac{7}{48}$
- D** $\frac{1}{3}$
- E** $\frac{3}{8}$
- F** $\frac{5}{8}$
- G** $\frac{1}{6}$
- H** $\frac{5}{12}$

- 13** A machine extrudes a solid cylindrical rod of plastic. The plastic flows out at a constant rate of $0.072\pi \text{ m}^3$ per hour. The diameter of the extruded rod is constant at 40 mm.

What is the linear speed at which the rod is produced, in cm/s?

- A** 0.05
- B** 0.5
- C** 1.25
- D** 5
- E** 75
- F** 300

- 14** A company purchases large quantities of three raw materials. The prices of these materials are p , q , and r dollars per tonne, respectively.

The company spends an equal amount of money on each of the three materials.

What is the average price, in dollars per tonne, that the company pays for all the materials purchased?

- A** $\frac{p+q+r}{3}$
B $\frac{3}{p+q+r}$
C $\frac{3pqr}{p+q+r}$
D $\frac{3pqr}{pq+qr+rp}$
E $\frac{pq+qr+rp}{3pqr}$
F $\frac{p^2+q^2+r^2}{p+q+r}$

- 15** A point P moves anticlockwise on a circle with equation $x^2 + y^2 = R^2$, where $R > 0$. It starts at the point $(R, 0)$ and moves with a constant period of T for one full revolution.

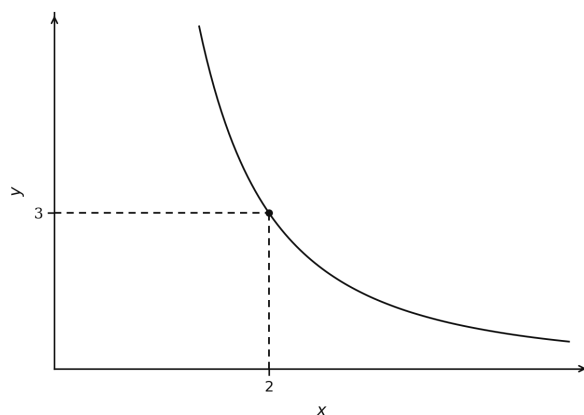
A second point, Q, starts at (R, H) , where $H > 0$, and moves vertically downwards with a constant speed v .

The two points start moving at the same time. Q reaches the x -axis at the exact same instant that P reaches the positive y -axis for the first time.

What is v in terms of H and T ?

- A** $\frac{4H}{T}$
B $\frac{2H}{T}$
C $\frac{H}{T}$
D $\frac{HT}{4}$
E $\frac{4R}{T}$
F $\frac{2H}{\pi T}$

- 16** The diagram shows the graph of y against x for $x > 0$, where y is inversely proportional to x^2 . The variable x is directly proportional to z^3 . When $z = 2$, $x = 4$. Which of the following expressions gives y in terms of z ?



- | | |
|----------|----------------------|
| A | $y = \frac{3}{z^6}$ |
| B | $y = \frac{24}{z^6}$ |
| C | $y = \frac{48}{z^6}$ |
| D | $y = \frac{48}{z^5}$ |
| E | $y = \frac{24}{z^3}$ |
| F | $y = \frac{48}{z^3}$ |
| G | $y = \frac{z^6}{48}$ |
| H | $y = \frac{z^6}{12}$ |

- 17 Two vertices of a regular hexagon are located at $(-1, 2)$ and $(3, 6)$. What is the difference between the areas of the largest and smallest possible regular hexagons that can be drawn with these points as two of their vertices?

A $4\sqrt{3}$
 B $12\sqrt{3}$
 C $24\sqrt{3}$
 D $32\sqrt{3}$
 E $36\sqrt{3}$
 F $48\sqrt{3}$
 G $60\sqrt{3}$
 H $64\sqrt{3}$

- 18 A set contains the integers from 1 to n , where $n \geq 2$.
 Two distinct integers are chosen at random from the set.
 The probability that their sum is odd is $\frac{12}{23}$.

What is the sum of all possible values of n ?

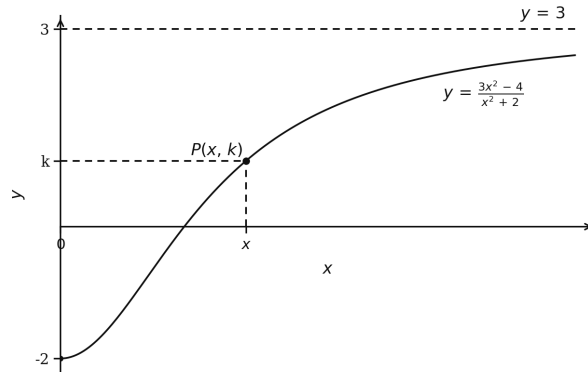
A 23
 B 24
 C 35
 D 46
 E 47
 F 48

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The diagram shows the graph of $y = \frac{3x^2 - 4}{x^2 + 2}$ for $x \geq 0$, with a horizontal asymptote at $y = 3$ and a y -intercept at $(0, -2)$.

A horizontal line $y = k$, where $-2 < k < 3$, intersects the curve at the point $P(x, k)$.

Which of the following is an expression for x in terms of k ?



- A** $x = \sqrt{\frac{2k + 4}{3 - k}}$
- B** $x = \sqrt{\frac{2k + 4}{3 + k}}$
- C** $x = \sqrt{\frac{4 - 2k}{3 - k}}$
- D** $x = \sqrt{\frac{4 - 2k}{3 + k}}$
- E** $x = \sqrt{\frac{k + 4}{3 - k}}$
- F** $x = \sqrt{\frac{3 - k}{2k + 4}}$

- 20** The table below shows the frequency distribution of a data set, where x is a positive integer.

Value	Frequency
1	4
2	x
3	7
4	$2x$
5	5

The mean of the data set is 0.2 greater than the median of the data set.

What is the value of x ?

- A** 2
 - B** 3
 - C** 4
 - D** 5
 - E** 6
- 21** A circle is tangent to the line $y = 0$ and the line $3x - 4y = 0$. The circle passes through the point $(6, 3)$.
What is the sum of the possible values for the radius of the circle?
- A** 3
 - B** $14/3$
 - C** 5
 - D** 15
 - E** 18

- 22** A dataset consists of n numbers, where n is an odd integer greater than 1. The mean, median, and range of the dataset are denoted by μ , m , and r respectively.

A single new number with value m is added to the dataset.

Let μ' , m' , and r' denote the mean, median, and range of the new dataset.

Which of the following statements must be true?

1. $m' = m$ 2. $r' = r$ 3. $\mu' = \mu$

- A** 1 only
B 2 only
C 3 only
D 1 and 2 only
E 1 and 3 only
F 2 and 3 only
G 1, 2 and 3
H None of the statements
- 23** The coefficients B and C of the quadratic equation $x^2 + Bx + C = 0$ are determined as follows. A fair coin is tossed two times, and B is the number of heads. A second fair coin is tossed one time, and C is the number of heads.

What is the probability that the equation has at least one real root?

- A** $\frac{1}{3}$
B $\frac{3}{8}$
C $\frac{1}{2}$
D $\frac{5}{8}$
E $\frac{2}{3}$
F $\frac{7}{8}$

- 24** A function F is constructed by composing four functions in a sequence:

$$F(x) = f_4(f_3(f_2(f_1(x))))$$

Each of the functions f_1, f_2, f_3, f_4 is chosen independently from a set containing 3 distinct functions.

Given that any two different sequences of choices result in a distinct function F , how many possible functions F can be constructed?

- A** 12
- B** 15
- C** 24
- D** 27
- E** 64
- F** 81

- 25** A function f is defined for $x > 0$ by

$$f(x) = a\sqrt{x} + \frac{b}{\sqrt{x}}$$

where a and b are constants.

Given that $f(1) = 16$ and $f(4) = 14$, what is the value of $f(16)$?

- A** 6
- B** 10
- C** 14
- D** 19
- E** 28

- 26** The positive variables p, q, r, k, x, y are related by the equations:

$$p = qr$$

$$r = \frac{kx}{y}$$

Given that $p = 6.0 \times 10^1$, $q = 3.0 \times 10^{-2}$, $k = 4.0 \times 10^{-5}$, and $y = 4.0 \times 10^{-8}$, what is the value of x ?

- A** 0.2
 - B** 0.5
 - C** 2.0
 - D** 20
 - E** 1.8×10^{-3}
 - F** 2.0×10^6
- 27** A circular sector has a total area of 18π . The area of the sector per unit of arc length is 3. What is the arc length of the sector?
- A** 3π
 - B** 6
 - C** 6π
 - D** 12π
 - E** 54π